

Private Query Refinement

KEYWORDS

Query refinement, diversity, provenance, ranking, top-k

PVLDB Reference Format:

. Private Query Refinement. PVLDB, 14(1): XXX-XXX, 2020.
doi:XX.XX/XXX.XX

PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at
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1 INTRODUCTION

(Alon) TBD

2 PRELIMINARIES

2.1 Databases, Queries, Constraints, and Histograms

(AG) TBD: define the model for a single table database, the query class that we support, the representation constraints from your paper. Citations, intuition, formal definition.

Our setting begins with a relational database that stores tuples - Each representing an individual, record, or entity. A user submits a structured query (typically a SP query) to retrieve information that meets certain conditions. However, in many analytical or decision-making scenarios, simply returning the query results may inadvertently reinforce bias or underrepresent certain subgroups. To reason about and control such effects, we use [1, 4, 8] to define how sensitive attributes, groups, and constraints are modeled within the database.

A **sensitive attribute** captures information that has ethical, social, or regulatory implications - such as race, gender, or age group. These attributes define **groups** of individuals whose presence or absence in the query output can meaningfully affect fairness or diversity. For example, in a candidate-ranking scenario, "women with low income" forms one such group. **Cardinality constraints** over these groups allow us to formalize statements such as "at least 30% of the selected candidates should be women", turning fairness or representational goals into concrete mathematical conditions. When the user's original query violates these constraints, we may search for a **query refinement** - a minimal modification of the query predicates designed to restore compliance while preserving the user's intent.

Definition 2.1 (Sensitive Attribute [8]). We define a *sensitive attribute* as a discrete-valued feature central to fairness and diversity, such as race or gender. While the model assumes a single sensitive

attribute, multiple attributes can be handled independently or combined into a composite attribute (e.g., Race $\times$ Gender). Continuous or high-cardinality attributes (e.g., age, salary) can be bucketized. We assume the dataset is partitioned by the sensitive attribute, meaning each individual is assigned exactly one value.

Definition 2.2 (Group [1, 4]). A *group* is a collection of tuples that share the same value(s) for one or more (categorical) attributes and is defined by a conjunction of conditions over values of the attributes. Its notation is $\bigwedge_{A \in \text{attrs}} A = \text{Value}_A$. For example, Gender = F $\wedge$ Income = Low.

Definition 2.3 (Cardinality constraint [5, 7]). Applying cardinality constraints on group memberships in query results has been proposed to mitigate bias and improve diversity [5, 7]. A cardinality constraint $\ell_G = n$ (or $u_G = n$) specifies a lower (or an upper) bound of n tuples belonging to a group G appearing within the result. For instance, in our running example, the constraint "at least 3 of the candidates are women", can be expressed as $\ell_{\text{Gender}=F} = 3$. Multiple cardinality constraints may be composed together, forming a constraint set that we denote by C .

Definition 2.4 (Constraints satisfaction [5, 7]). Given a database D , a SPJ query Q and a set of cardinality constraints C , we say that Q *satisfies the constraints* iff every constraint $c \in C$ is satisfied by the set of tuples $Q(D)$.

Definition 2.5 (refinements [6]). Given a query Q , a refinement of Q modifies its selection predicates. A numerical predicate $A \diamond C \in \text{Num}(Q)$ is a modification to the value of C . For categorical predicates $\bigvee_{c \in C} A = c \in \text{Cat}(Q)$, a refinement is done by adding and/or removing predicates from the set of values C . We say that a query Q' is a *refinement* of query Q if Q' is obtained from Q by refining some predicates of Q .

Definition 2.6 (Histogram [?]). Given a database D , and a categorical attribute A , we say that the *histogram* defined by them is the distribution function $H_{D,A} : \text{Dom}(A, D) \rightarrow \mathbb{N}$ of the values of A in the tuples of D . This definition can be extended to support numeric attribute using bucketing.

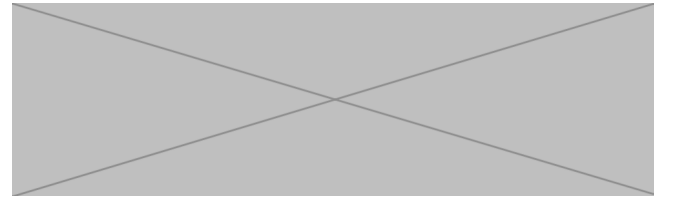


Figure 1: Example Placeholder

2.2 Differential Privacy

(AG) TBD: sensitivity, DP definition, laplace mechanism, exponential mechanism, relevant theorems (the exp mechanism is DP, composition theorems...)

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Proceedings of the VLDB Endowment, Vol. 14, No. 1 ISSN 2150-8097.
doi:XX.XX/XXX.XX

Definition 2.7 (Neighbors Dataset [3]). We say that two datasets D, D' are neighbors iff $|D \Delta D'| = 1$. We denote it by $D \sim D'$.

Definition 2.8 (Differential Privacy [2, 3]). An algorithm $\mathcal{M}$ is ϵ -differentially private (or ϵ -DP) iff for all $S \subseteq \text{Range}(\mathcal{M})$ and for all D, D' neighbors datasets: $\Pr[\mathcal{M}(D) \in S] \leq \exp(\epsilon) \Pr[\mathcal{M}(D') \in S]$, where the probability space is over the coin flips of the mechanism $\mathcal{M}$.

Definition 2.9 (Global Sensitivity [3]). The global-sensitivity (Δ) of a function $f : D \rightarrow \mathbb{R}$ is:

$$\Delta f = \max_{D \sim D' \subseteq D} |f(D) - f(D')|$$

Definition 2.10 (Laplace Distribution [3]). The Laplace Distribution (centered at 0) with scale b is the distribution with probability density function:

$$\text{Laplace}(x|b) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right).$$

We will sometimes write $\text{Laplace}(b)$ to denote a Laplace Distribution with scale b , and also write $\text{Laplace}(b)$ to denote a random variable $X \sim \text{Laplace}(b)$.

Definition 2.11 (The Exponential Mechanism [3]). Given a private database D , set of possible outcomes Y , utility function $f : Y \times D \rightarrow \mathbb{R}$ with global sensitivity Δ , and a privacy budget ϵ , The exponential mechanism (EM) selects and outputs an element $y \in Y$ with probability proportional to $\exp(\frac{\epsilon f(y, D)}{2\Delta})$. It has been proven in [3] that the exponential mechanism is ϵ -DP

3 MODEL

3.1 Private query refinement model

(AG) define the framework: user sends a query to the trusted server, the server refines it,...

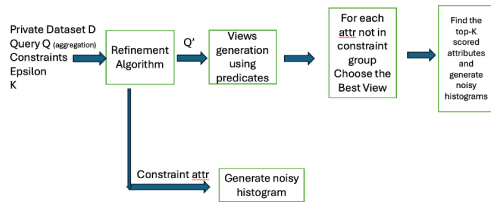


Figure 2: An illustration of the framework

(Alon) I think I should remove the "Views generation" block from this scheme and reveal it during the Algorithm section

We expect to get the following arguments from the user:

- (1) A private database D ;
- (2) A SP aggregate query Q ;
- (3) A set of cardinality constraints C ;
- (4) A refinement mechanism $\mathcal{M}(Q, C)$;
- (5) A differential privacy budget $\epsilon > 0$;
- (6) The number of explanations to release $k \in \mathbb{N}$.

We aim to reveal a privacy-preserving explanation for the query refinement that satisfies the diversity constraints, while providing differentially private explanations for how and why the query was

modified. To achieve this, we give some differentially private histograms that highlight changes in the data distribution caused by the refinement, offering insight into the impact of the modified query predicates.

3.2 Query difference explanation

(AG) Define our model for explanations for query difference: COH, QOH, Overall quality function for a histogram explanation.

Definition 3.1 (Noisy Histogram). Given $\epsilon < 1$, we say that $H_{D,A}^{\hat{}}$ is a noisy histogram iff $H_{D,A}^{\hat{}}(v) = H_{D,A}(v) + \text{Laplace}(1/\epsilon)$.

Definition 3.2 (Constraint Oriented Histogram (COH)). Given a database D and a set of cardinality constraints C , a constraint oriented histogram is an histogram $H_{D,A}$ such that A appears in one or more groups in the constraints in C .

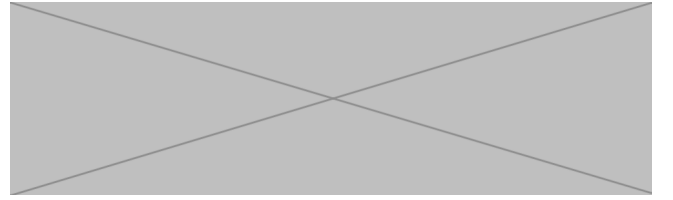


Figure 3: Example Placeholder

Definition 3.3 (Query Oriented Histogram (QOH)). Given a database D and a SPJ query Q , a query oriented histogram is an histogram $H_{Q(D),A}$.

Definition 3.4 (Refinement Predicate Explanation Candidates). Given a database D , a SPJ query Q , set of cardinality constraints C , Q' a refinement of Q which satisfies the constraints and an attribute $A \in \text{Attr}(D)$, we say that the group of refinement predicate explanation candidates for Q' and A is the following set of histograms: $H_{D,Q}(Q', A) = \{(H_{Q'(D'),A}, H_{Q(D'),A}) | D' \subseteq D\}$

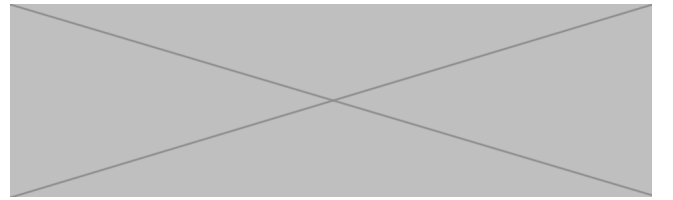


Figure 4: Example Placeholder

(Alon) Find different symbol(Alon) Convert to definition of single candidate

Definition 3.5 (DP-Preserving Predicate Explanation Candidate). Given the same setup, the group of DP-preserving predicate explanation candidates is the set of noisy histograms which is defined as follows:

$$\hat{H}_Q(Q', A) = \{(\hat{H}_{Q'(D'),A}, \hat{H}_{Q(D'),A}) | D' \subseteq D\}$$

(Alon) Add the definition of the specific utility, without context

Definition 3.6 (Best (Alon) Change to optimal DP-Preserving Predicate Explanation). Given a database D , a SPJ query Q , a refinement Q' , an attribute $A \in \text{Attr}(D)$ and a utility function $U(Q, A)$, the *best DP-preserving predicate explanation* is the DP-preserving predicate explanation candidate

$$(H_1, H_2) = \arg \max_{A' \in \text{Attr}(D)} U(D, Q, Q', A').$$

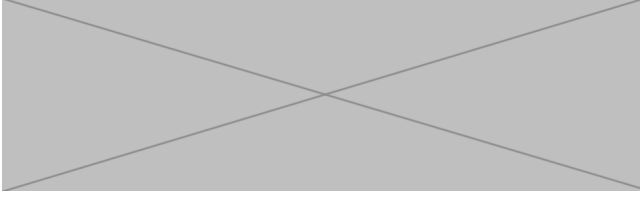


Figure 5: Example Placeholder

3.3 Problem definition

(AG) Give an intuitive explanation and motivation for the problem definition and formally define it using the definitions from the previous sections.

Let:

- (1) D be a private database;
- (2) Q be a user-submitted SPJ aggregate query;
- (3) C be a set of *cardinality constraints* over groups defined by values of one or more sensitive attributes;
- (4) Q' be a refinement of Q which satisfies the constraints C .
- (5) $\epsilon > 0$ be the differential privacy budget;
- (6) $k \in \mathbb{N}$ be the number of explanations to release.

We aim to compute:

- (1) A set of *noisy constraint-oriented histograms*:

$$\hat{H}_{Q',A}(v) = H_{Q',A}(v) + \text{Lap}(1/\epsilon), \quad \forall A \in \text{attrs}(C), v \in \text{Dom}(A)$$

- (2) A set of k *differentially private explanation candidates*, selected to maximize histogram divergence between Q and Q' .

$$k\text{-ArgMax} \{Utility\}, \quad A \in \text{attrs}(D)$$

To evaluate and select meaningful explanations for query refinements, we define a utility function that incorporates three key components: *interestingness*, *representativeness*, and *coverage*. Each explanation is based on a view predicate p over a subset of attributes $A_1, \dots, A_m$, **(Alon) Formal definition** applied to a private dataset D , and compared between the original query Q and its refined version Q' . For a chosen attribute A , we define: **(Alon) Add references to previous works, and extend the explanations for the chosen metrics**

- (1) the **interestingness** score quantifies the absolute difference in the conditional distributions of A between $Q(p(D))$ and $Q'(p(D))$, aggregated over discrete buckets of A .

$$\sum_{V \in \text{buckets}(D[A])} |\Pr(t[A] = V \mid t \in Q(p(D))) - \Pr(t[A] = V \mid t \in Q'(p(D)))|$$

- (2) The **representativeness** score measures how closely the distribution of A within the view $p(D)$ matches its distribution in the full dataset D , encouraging views that generalize well.

$$1 - \sum_{V \in \text{buckets}(D[A])} |\Pr(t[A] = V \mid t \in D) - \Pr(t[A] = V \mid t \in p(D))|$$

- (3) The **Coverage** is defined as the product of the sizes of $p(D)$, $Q(p(D))$, and $Q'(p(D))$, promoting views that are statistically robust.

$$|p(D)| \times |Q(p(D))| \times |Q'(p(D))|$$

The final utility score is computed as the product of Interestingness, Representativeness and Coverage:

$$\text{Utility} = \text{Interestingness} \times \text{Representativeness} \times \text{Coverage}$$

(Alon) Add the connection between the theoretical metrics to the metrics we choose

4 ALGORITHM

(AG) Explain the quality function and why it is a good proxy for the optimization function that you presented in Section 3, add proposition and proof for its sensitivity, add the algorithms (use the one-shot-top-k mechanism which is equivalent to k times the exponential mechanism, see [9]), write the pseudo code, add a textual explanation for the code, and a proposition and proof for the algorithm being DP.

(AG) Challenges: (1) generating all possible views is impossible so considering only views that are expressed by predicates (common in explanation works), (2) the optimization function has high sensitivity and high computational cost even with views generated by predicates so we define a proxy function with limited sensitivity that can be easily computed.

Designing a practical algorithm for private query-refinement explanations requires addressing two central challenges: the enormous combinatorial space of potential views and the high sensitivity of the optimization objective defined in Section 3. Directly optimizing this objective is computationally infeasible and would yield poor differential privacy guarantees. To overcome these obstacles, our approach introduces a carefully crafted proxy quality function with bounded sensitivity, enabling efficient evaluation under differential privacy. We then propose the following framework for the solution of this problem as shown in figure 6:

- (1) Given a private database D , a query Q , and a constraints set C , we run a refinement mechanism and get a refinement Q' that satisfies the constraints.
- (2) Generate a set of predicate views of the data that will form the basis on top of which we will create the histograms.
- (3) Filter out views that are "less useful" in terms of explaining trends. According to a usefulness measure which is a proxy function with low sensitivity (e.g. views that are very small,

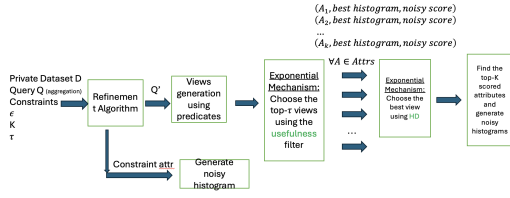


Figure 6: An illustration of the framework

or that are very different from the original database in the distribution of the features).

- (4) For each attribute, find the view that best reflects the change between the results of $Q(D)$ and $Q'(D)$ for this attribute, in terms of the difference in the one-way marginals of this attribute.
- (5) Release the best- k noisy histograms of the different attributes from the set of tuples (Attribute, Best View, Noisy Score) we collected in the previous step.

4.1 Proxy Measures

We introduce the proxy functions for the Usefulness and the one-way marginals (Histograms Distance) and proof they are sensitive bounded.

Definition 4.1 (Histogram Distance). Given a dataset D , a view predicate p , an attribute A , and two queries Q and Q' , the *Histogram Distance* (HD) is defined as:

$$HD = \sum_{V \in \text{buckets}(D[A])} \frac{|H_{Q(p(D)),A}(V) - H_{Q'(p(D)),A}(V)|}{H_{p(D),A}(V)}, \quad (1)$$

where $H_{Q(p(D)),A}(V)$ denotes the count of records in bucket V for attribute A in the result of Q applied to $p(D)$, and $H_{p(D),A}(V)$ is the total count in bucket V for $p(D)$.

PROPOSITION 4.2. The global sensitivity of HD as defined in (1) is at most 1, and its range is $0 \leq HD(D) \leq |p(D)|$.

PROOF. Let D, D' be neighboring datasets such that $||D'| - |D|| = 1$ and $D' = D \cup \{t\}$, where t is a single tuple. Let $t_A = V'$ denote the bucket value for attribute A of tuple t .

The global sensitivity of HD is:

$$GS_{HD} = \max_{\substack{D, D' \\ D \sim D'}} |HD(D) - HD(D')|, \quad (2)$$

where $D \sim D'$ denotes that D and D' are neighboring datasets.

From (1), we have:

$$\begin{aligned} & |HD(D) - HD(D')| \\ &= \left| \sum_{V \in \text{buckets}(D'[A])} \frac{|H_{Q(p(D)),A}(V) - H_{Q'(p(D)),A}(V)|}{H_{p(D),A}(V)} \right. \\ &\quad \left. - \sum_{V \in \text{buckets}(D'[A])} \frac{|H_{Q(p(D')),A}(V) - H_{Q'(p(D')),A}(V)|}{H_{p(D'),A}(V)} \right|. \quad (3) \end{aligned}$$

We partition the set of buckets into two disjoint subsets:

$$\text{buckets}(D'[A]) = (\text{buckets}(D'[A]) \setminus \{V'\}) \cup \{V'\}.$$

Case 1: $V \neq V'$. For all $V \neq V'$, the numerator in each term is unchanged between D and D' , while the denominator changes by at most 1 because the tuple t is not assigned to V . Thus:

$$\sum_{V \neq V' \in \text{buckets}(D'[A])} \left| \frac{H_{Q(p(D)),A}(V)}{H_{p(D),A}(V)} - \frac{H_{Q(p(D')),A}(V)}{H_{p(D'),A}(V)} \right| \leq 1.$$

Case 2: $V = V'$. Here, both numerator and denominator may change, but by the neighboring dataset property, each changes by at most 1. Therefore:

$$\begin{aligned} & \left| \frac{H_{Q(p(D)),A}(V') - H_{Q'(p(D)),A}(V')}{H_{p(D),A}(V')} \right| \\ & - \left| \frac{H_{Q(p(D')),A}(V') - H_{Q'(p(D')),A}(V')}{H_{p(D'),A}(V')} \right| \leq 1. \quad (4) \end{aligned}$$

Since only one bucket V' can change between D and D' , and the change in that bucket's contribution is bounded by 1, it follows that:

$$|HD(D) - HD(D')| \leq 1.$$

Taking the maximum over all neighboring datasets yields:

$$GS_{HD} \leq 1.$$

□

Definition 4.3 (Similarity Penalty). Let D be a dataset, $\mathcal{A}$ a set of attributes and A^* an attribute. We define the Similarity Penalty of A^* with respect to $D, \mathcal{A}$ as following:

$$\text{SimilarityPenalty}(A^*, \mathcal{A}, D) = \text{TVD}(D[A^*], D[\mathcal{A}]) \quad (5)$$

Definition 4.4 (Usefulness Index). Let D be a dataset, p a view predicate, A an attribute, and Q_1, Q_2 two queries. We define the coverage index as following:

$$\text{Coverage}(p, Q_1, Q_2, D) = \frac{|Q_1(p(D))| + |Q_2(p(D))|}{|D|} \quad (6)$$

Let $\text{Representativeness}(p, D) := \text{TVD}(D, p(D))$ denote the total variation distance as a Representativeness index. The *Usefulness index* is defined as

$$\text{Usefulness}(p, Q_1, Q_2, D) = \text{Representativeness}(p, D) \cdot \text{Coverage}(p, Q_1, Q_2, D) \quad (7)$$

PROPOSITION 4.5 (GLOBAL SENSITIVITY OF Usefulness). Using previous works that showed: **(Alon) Add reference**

- (1) $\text{TVD}(\cdot) \in [0, 1]$
- (2) $GS_{\text{TVD}} = 1/|D|$,

Then

$$\Delta_{\text{Usefulness}} = \max_{D \sim D'} |U(D) - U(D')| \leq 4.$$

and its range is bounded by $0 \leq \text{Usefulness}(D) \leq \frac{2|p(D)|^2}{|D|}$.

PROOF. Let D, D' be neighboring datasets with $D' = D \cup \{t\}$. We denote:

$$x = \text{TVD}(p, D), \quad y = \text{Coverage}(p, Q, Q', D),$$

and similarly

$$x' = \text{TVD}(p, D'), \quad y' = \text{Coverage}(p, Q, Q', D')$$

Therefore, we have:

$$\begin{aligned} |Usefulness(p, Q, Q', D) - Usefulness(p, Q, Q', D')| &= |xy - x'y'| \\ &= |xy - x'y + x'y - x'y'| \leq |x - x'| |y| + |y - y'| |x'| \end{aligned} \quad (8)$$

(i) *First term bound.* We can bound $|y| \leq 2$:

$$y = \frac{|Q(p(D))| + |Q'(p(D))|}{|D|} \leq \frac{|D| + |D|}{|D|} = 2,$$

Therefore, we get:

$$|x - x'| |y| \leq |x - x'| \cdot 2 \leq 2 \cdot \text{GSTVD} = 2/|D|. \quad (9)$$

(ii) *Second term bound.*

$$\begin{aligned} |y - y'| &= \left| \frac{|Q(p(D))| + |Q'(p(D))|}{|D|} - \frac{|Q(p(D'))| + |Q'(p(D'))|}{|D'|} \right| \\ &= \left| \frac{|Q(p(D))| + |Q'(p(D))|}{|D|} - \frac{|Q(p(D'))| + |Q'(p(D'))|}{|D| + 1} \right| \\ &\leq \left| \frac{|Q(p(D'))|}{|D| + 1} - \frac{|Q(p(D))|}{|D|} \right| + \left| \frac{|Q'(p(D'))|}{|D| + 1} - \frac{|Q'(p(D))|}{|D|} \right| \\ &= \left| \frac{|Q(p(D'))| \cdot |D| - |Q(p(D))| \cdot (|D| + 1)}{|D|(|D| + 1)} \right| + \\ &\quad \left| \frac{|Q'(p(D'))| \cdot |D| - |Q'(p(D))| \cdot (|D| + 1)}{|D|(|D| + 1)} \right| \\ &= \left| \frac{|D| \cdot (|Q(p(D'))| - |Q(p(D))|) - |Q(p(D))|}{|D|(|D| + 1)} \right| + \\ &\quad \left| \frac{|D| \cdot (|Q'(p(D'))| - |Q'(p(D))|) - |Q'(p(D))|}{|D|(|D| + 1)} \right| \\ &= \left| \frac{|Q(p(D'))| - |Q(p(D))|}{|D| + 1} - \frac{|Q(p(D))|}{|D|(|D| + 1)} \right| + \\ &\quad \left| \frac{|Q'(p(D'))| - |Q'(p(D))|}{|D| + 1} - \frac{|Q'(p(D))|}{|D|(|D| + 1)} \right| \\ &\leq \left| \frac{1}{|D| + 1} + \frac{|Q(p(D))|}{|D|(|D| + 1)} \right| + \left| \frac{1}{|D| + 1} + \frac{|Q'(p(D))|}{|D|(|D| + 1)} \right| \\ &\leq 1 + 1 = 2. \end{aligned} \quad (10)$$

Therefore:

$$|y - y'| |x'| \leq 2 \cdot 1 = 2. \quad (11)$$

(iii) *Combine.* From (8), (9) and (11):

$$|Usefulness(p, Q, Q', D) - Usefulness(p, Q, Q', D')| \leq 2 + 2 = 4. \quad (12)$$

Therefore, $\text{GS}_{Usefulness} \leq 4$. $\square$

4.2 Solving the Private Query-Refinement Explanations Problem

In this subsection we introduce our algorithm for solving the Private Query-Refinement Explanations problem. We first introduce the overall algorithm which implements the framework of figure 6 and then the auxiliary algorithms it uses. Then we show that each algorithm is $\epsilon - DP$ and conclude that the overall algorithm is also $\epsilon - DP$.

Algorithm 1 Private-Queries-Diff-Top-K-Explanation

Require: Private dataset D , SPJ Queries Q_1, Q_2 , Set of available attributes $Attrs$, Set of predicates views $\mathcal{P}$, threshold τ , k , Privacy budget ϵ

- 1: $\hat{\mathcal{P}} \leftarrow \text{One-Shot Top-}\tau(\mathcal{P}, D, \frac{\epsilon}{2})$ **{(Alon) todo}**
- 2: $Candidates \leftarrow \emptyset$
- 3: **for** each $P \in \hat{\mathcal{P}}$ **do**
- 4: **for** each $A_i \in Attrs$ **do**
- 5: $C_{P,A_i} \leftarrow (H_{Q_1(P),A_i}, H_{Q_2(P),A_i})$
- 6: $Candidates \leftarrow Candidates \cup \{C_{P,A_i}\}$
- 7: **end for**
- 8: **end for**
- 9: $Explanations \leftarrow \text{Find-Top-k-Explanations}(Attrs, Candidates, Q_1, Q_2, k, \frac{\epsilon}{2})$
- 10: **return** $Explanations$

Explanation. Algorithm 1 orchestrates the full pipeline. It refines the original query using the given refinement mechanism, generates predicate views from attribute pairs and domain values, uses a one-shot top- τ mechanism to privately select the most useful views, invokes Algorithm ?? to build noisy histograms for each view, and finally calls Algorithm 2 to return the top- k privacy-preserving explanations.

Algorithm 2 Find-Top-k-Explanations

Require: Private dataset D , SPJ Queries Q_1, Q_2 , Set of available attributes $Attrs$, Set of candidates C , k , Privacy budget ϵ

- 1: $R[1 \dots |C|] \leftarrow \text{NULL}, \dots, \text{NULL}$
- 2: $ChosenAttrs \leftarrow \emptyset$
- 3: **for** $i \in \{1 \dots k\}$ **do**
- 4: $H_i, A_i \leftarrow \text{EM}(C, \text{HD}(D, Q_1, Q_2) \times \text{DiversityScore}(ChosenAttrs))$
- 5: $H_i \leftarrow \text{Add-Histogram-Noise}(H_i, \epsilon/k)$
- 6: $R[i] \leftarrow H_i$
- 7: $ChosenAttrs \leftarrow ChosenAttrs \cup \{A_i\}$
- 8: $C \leftarrow C - H_i$
- 9: **end for**
- 10: **return** $R_1, \dots, R_k$

Explanation. Algorithm 2 selects the k most informative and privacy-preserving explanations from the candidate set. For each attribute A_i , the algorithm applies the exponential mechanism (EM) to the subset of candidates corresponding to that attribute. We assume the EM returns both the noisy score and the chosen histogram H_i , ensuring that the selection process satisfies the privacy budget ϵ . These noisy scores are then used to rank all candidate explanations. Finally, the algorithm sorts the results by descending noisy score and outputs the top- k explanations, balancing utility and privacy guarantees.

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